ACCESSIBILITY · CHROME EXTENSION

iBrowse

Hear any website. Explore it by **mouse**, **touch**, **keyboard** and **voice**, built for blind and low-vision people.

Free and open source · github.com/srksourabh/ibrowse



WHY IT MATTERS

The web is built to be looked at.

Layout tells sighted people where things are: the menu on the left, the offers on the right, the button under the picture. Without sight, that map is missing.

2.2 billion

people live with a near or distance vision impairment (WHO, World report on vision, 2019).

Where?

Screen readers read in order. iBrowse adds *where*: what is under your hand, and what is on each side.

Any site

No changes to websites needed. iBrowse reads the page you are on.

HOW IT WORKS · IN FIVE STEPS

Turn it on, listen, explore, talk, act.



Turn on

Alt+Shift+E



Briefing

panel by panel



Explore

hover, touch, wheel



Talk

hold Alt+Shift+V



Act

click, scroll, read

Nothing happens until you turn Explore Mode on. Off, the page behaves exactly as usual.

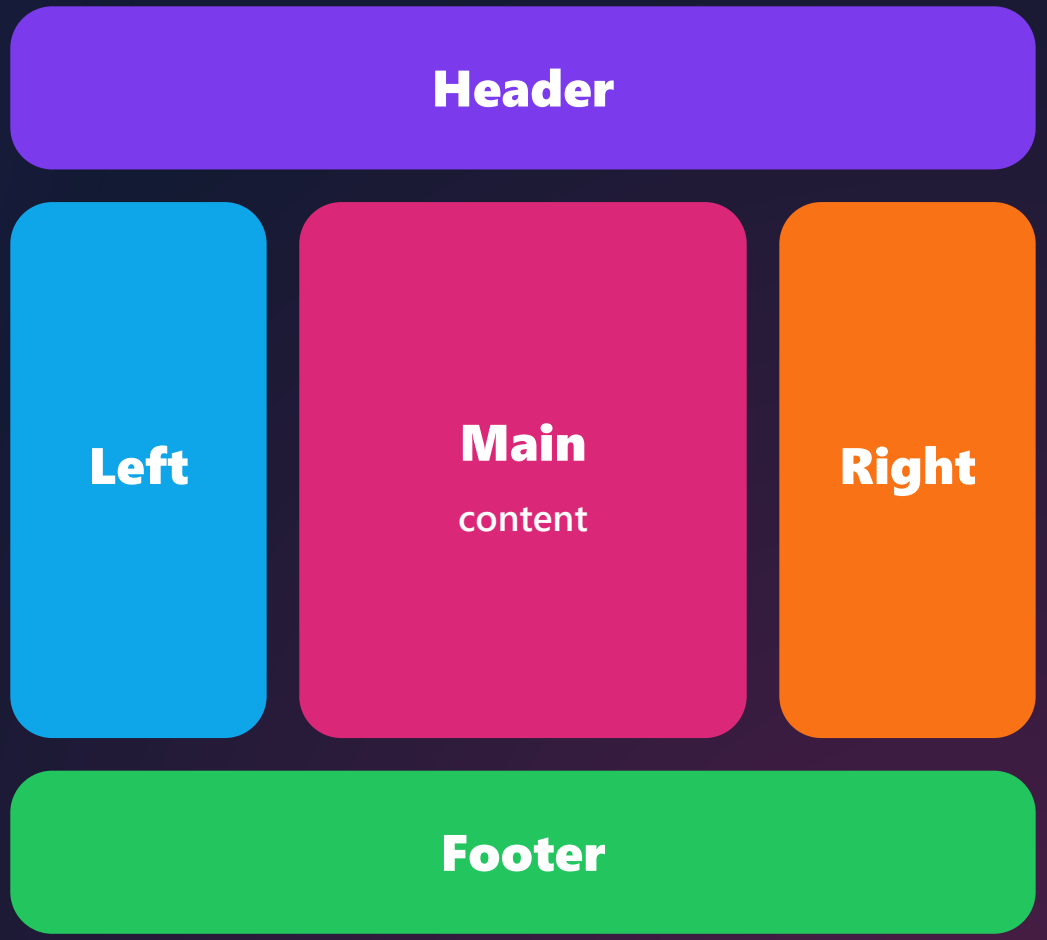
PAGE BRIEFING

A map of the page, spoken.

iBrowse finds the header, the side panels, the main content and the footer, from landmarks or from the layout itself, and says what each one holds.

IBROWSE SAYS

Header: search box, 4 links. Left panel, Categories: 3 links. Main content, Cherry tomatoes: 2 paragraphs, 1 picture, 2 buttons. Right panel, Offers: 2 links. Want me to read it?



EXPLORE BY POINTER AND TOUCH

Whatever is under your hand, named.

Hover

"Add to basket — Button, Clickable." Speech stops the moment you move on.

Sound in space

Short earcons mark links, buttons, text and edges, panned left or right to match where they are.

Touch

Drag a finger to explore, double-tap to activate, and hear an edge tone at the side of the screen.

Rest to ask

Rest on something and iBrowse asks "What would you like to do?". Answer out loud.

READING

Click to read. Scroll word by word.

Click a paragraph to hear it. Over text, the mouse wheel steps one word at a time, Shift + wheel spells a word, and Alt+Shift+R reads on from where you are.

Sweet, easy to **grow**
in a sunny pot.



VOICE CONVERSATION

Can't click? Say it.

"Click"
"Click add to basket"
"Scroll down"
"Scroll up"
"What is on the left?"
"Read the page"
"Next page"
"Go back"
"Find the search box"
"Where am I?"
"Stop"
"Help"

Commands are understood on your computer. Anything else is a question, answered from the page if you turn on AI.

YOU SAY
Click buy now.

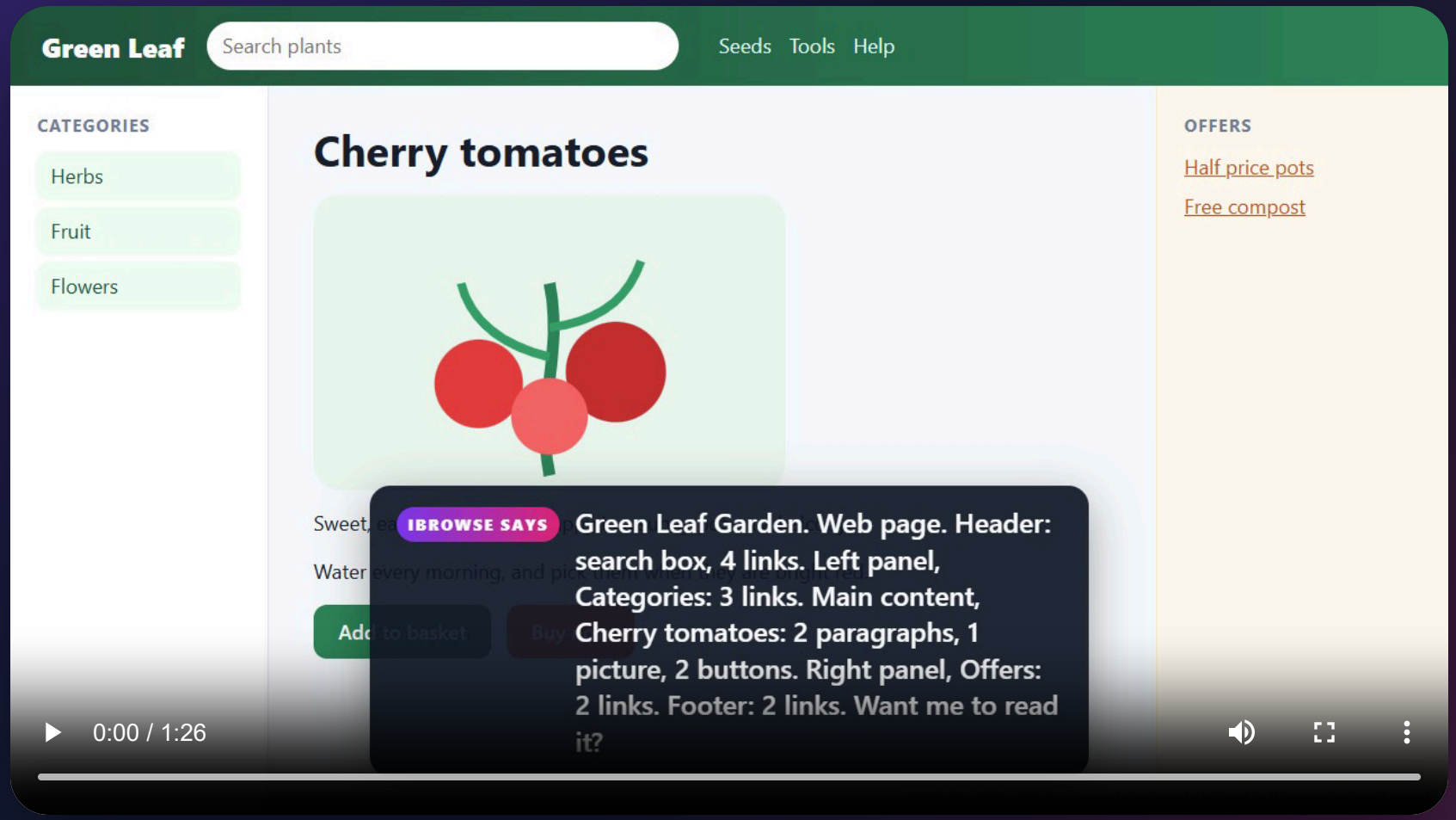
IBROWSE SAYS
Buy now. Say yes to press it, or no.

YOU SAY
Yes.

Anything that can't be taken back (paying, sending, deleting, agreeing) waits for a spoken yes, and a harmless label alone can't hide it.

SEE IT · AND HEAR IT

iBrowse in action



A recording of iBrowse on a made-up shop page, with captions. The user's voice commands are simulated, and both voices are synthesised.

PLAYS WELL WITH OTHERS

Your screen reader, your hardware.

Companion mode

NVDA speaks for iBrowse, so you hear one voice. Checked with NVDA: the briefing and hover descriptions were spoken once, and push to talk worked while NVDA ran.

Feel it

Every sound has a matching touch effect: a rumbling game controller, or a USB haptic device described in JSON.

Open SDK

Hardware makers get a schema, an example profile, an example driver and conformance checks.

PRIVATE BY DEFAULT

It stays on your computer.

No servers, no accounts, no analytics. Optional Google Gemini features (a natural voice, AI summaries and answers, image descriptions) use your own key. iBrowse says what will be sent before you allow it.

Never sent: form values, passwords, cookies, browsing history, or anything while Explore Mode is off.



UNDER THE HOOD

A small, tested Chrome extension

In every page

- briefing and panels
- hover, touch, reading
- forms and live changes
- voice commands

content script, every frame

Background worker

- the computer's voices
- settings, toolbar, keys
- the Gemini key, locked away
- haptic devices

Speech recognition

hidden page, only while listening

Google Gemini

optional, your own key

QUALITY

Tested like it matters.

1,094

unit tests, about 98% of lines covered

99 × 5

end-to-end tests in real Chromium, five runs in a row for every speech change

398 / 444

official W3C accessible-name tests matched

Real voice

A person said "Scroll down" into a real microphone, and iBrowse scrolled

Real NVDA

Companion mode checked against NVDA's own speech log

GET STARTED · HELP SHAPE IT

Try it, then tell us.

1

Build

`npm install` then `npm run build`

2

Load

chrome://extensions, Developer mode, Load unpacked, choose `.output/chrome-mv3`

3

Explore

A spoken tour opens. Then press `Alt+Shift+E` on any page.

Next: a closed beta with blind testers, then the Chrome Web Store. Free and open source under the MIT licence.

github.com/srksourabh/ibrowse